

Nikhil Singh

Research Engineer, AtheroPoint LLC

 Portfolio  Github  LinkedIn  Email

Education

July 2021	Guru Gobind Singh Indraprastha University	Delhi, India
July 2025	B.Tech., Electronics & Communication Engineering GPA: 8.3/10 Research assistant under Dr. Arun Kumar Dubey .	

Experience

May 2024	AtheroPoint LLC	California, USA
Present	Research Engineer Principal Investigator: Prof. Jasjit S. Suri, AtheroPoint Built (a) UNet-transformer/Mamba models for carotid ultrasound segmentation; (b) automated cIMT and plaque-burden measurement; (c) validated against expert tracings across 5K+ annotated scans; (d) deployed FastAPI/AWS inference services powering AtheroEdge 4.0.	
Dec 2024	Green Rider Technologies LLP	New Delhi, India
Feb 2025	Python Developer, LLM Fine-Tuning Curated 200K+ EN/HI/ES records and LoRA-tuned Gemini 1.0 Pro and LLaMA 2, with automated benchmarking across MMLU, GSM8K, and prompt-injection probes.	

Research Publications & Patents

Complete list at  [Google Scholar](#)

- [J] **Attention/Transformer-Based AI Models for Carotid Segmentation and IMT/Plaque Area Measurements in Japanese Ultrasound Scans**
Nikhil Singh, A. K. Dubey, ..., J. S. Suri [\[View\]](#)
Measurement (Elsevier), 2026 [Q1, IF 6.1]
- [J] **Twelve Fused Models: Four Transformer-Based Tuners on Three UNet Architectures for Carotid Wall Segmentation and Plaque Burden/IMT Measurement**
Nikhil Singh, ..., J. S. Suri [\[View\]](#)
Diagnostics (MDPI), 16(16):2538, 2026 [Q1, IF 3.8]
- [J] **A Comparative Study of State-of-the-Art AI Models for Automated Plaque Measurement in a Carotid Ultrasound Japanese Cohort**
Nikhil Singh, ..., J. S. Suri
Cardiovascular Diagnosis and Therapy (AME), 2026 (Major Revision) [Q1, IF 2.2]
- [J] **Y-Net-Based Wall Thickness and Plaque Burden Measurements in Ultrasound Scans with Low, Moderate, and High-Risk Patients**
Z. Ruzsa, Nikhil Singh, ..., J. S. Suri
Scientific Reports (Nature), 2026 (Under Review) [Q1, IF 4.9]
- [P] **Method and Apparatus for Cardiovascular Disease Risk Assessment**
J. S. Suri, Nikhil Singh (Co-Inventor) Appln No: 19/738,450
U.S. Patent filed with USPTO, assigned to AtheroPoint LLC [Jun 2026]
- [P] **Method and Apparatus for Cardiovascular Risk Assessment Using an Ensemble of Deep Neural Networks**
J. S. Suri, Nikhil Singh
U.S. Patent filed with USPTO, assigned to AtheroPoint LLC [Sep 2026]

References

Dr. Jasjit S. Suri
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Dr. Luca Saba
Professor of Radiology
University of Cagliari, Italy
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Dr. Arun Kumar Dubey
Associate Professor
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